5.14 Classwork: Logarithms and Exponentials — Markscheme

1. [Maximum mark: 5]

   Consider the curve with equation $y = (2x-1)e^{kx}$, where $x \in \mathbb{R}$ and $k \in \mathbb{R}$. The tangent to the curve at the point where $x = 1$ is parallel to the line $y = 5xe^k$. Find the value of k.

   Solution:

   Product rule: **(M1)**

   $\frac{dy}{dx} = 2 \cdot e^{kx} + (2x-1) \cdot ke^{kx} = e^{kx}(2kx - k + 2)$ **(A1)**

   At $x = 1$:

   $\frac{dy}{dx} = e^k(k+2)$ **(A1)**

   The line $y = 5xe^k$ has slope $5e^k$.

   Setting equal: **(M1)**

   $e^k(k+2) = 5e^k$

   $k + 2 = 5$

   $k = 3$ **(A1)**

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2. [Maximum mark: 9]

   Consider the function f defined by $f(x) = \ln(x^2 - 16)$ for $x > 4$.

   (a) Find the exact value of a. [3]

   Solution:

   $f(a) = 0 \implies \ln(a^2 - 16) = 0$ **(M1)**

   $a^2 - 16 = 1$ **(A1)**

   $a^2 = 17 \implies a = \sqrt{17}$ **(A1)**

   (b) Given that the gradient of L is $\frac{1}{3}$, find the x-coordinate of B. [6]

   Solution:

   $f'(x) = \frac{2x}{x^2 - 16}$ **(M1)(A1)**

   Setting $f'(x) = \frac{1}{3}$: **(M1)**

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$$\frac{2x}{x^2-16}=\frac{1}{3}$$

$6x = x^2 - 16$ **(A1)**

$x^2 - 6x - 16 = 0$

$(x-8)(x+2) = 0$ **(A1)**

$x = 8 \quad (\text{since } x > 4)$ **(A1)**

3. [Maximum mark: 14]

Let $g(x) = p^x + q$, for x, p, $q \in \mathbb{R}$, $p > 1$. The point $A(0, a)$ lies on the graph of g. Let $f(x) = g^{-1}(x)$.

(a) Write down the coordinates of B. [2]

Solution:

$g(0) = p^0 + q = 1 + q$, so $a = q + 1$ **(A1)**

B is the reflection of $A(0, a)$ in $y = x$, so $B = (a, 0)$ **(A1)**

(b) Given that $f'(a) = \dfrac{1}{\ln p}$, find the equation of L_1 in terms of x, p and q. [5]

Solution:

L_1 passes through $B(a, 0)$ with gradient $\dfrac{1}{\ln p}$ **(M1)**

$y - 0 = \dfrac{1}{\ln p}(x - a)$ **(A1)**

Substituting $a = q + 1$: **(M1)(A1)**

$y = \dfrac{1}{\ln p}(x - q - 1)$ **(A1)**

(c) Find the equation of L_1 in terms of x. [7]

Solution:

Finding p:

The gradient of the normal to g at A is $\dfrac{1}{\ln \frac{1}{3}} = \dfrac{1}{-\ln 3} = -\dfrac{1}{\ln 3}$ **(M1)**

Since (gradient of tangent) $\times$ (gradient of normal) $= -1$:

$(\ln p) \cdot \left(-\dfrac{1}{\ln 3}\right) = -1$ **(A1)**

$\ln p = \ln 3 \implies p = 3$ **(A1)**

Finding q:

L_2 passes through $(-2, -2)$:

$-2 = (\ln 3)(-2) + q + 1$ **(A1)**

$q = -3 + 2\ln 3$ **(A1)**

Finding L_1:

$a = q + 1 = -2 + 2\ln 3$ **(A1)**

$$L_1 :\ y = \frac{1}{\ln 3}(x - 2\ln 3 + 2) \qquad \textbf{(A1)}$$